

Design and Experimental Demonstration of a Conduction Cooling System for an S-band High-gradient Cryogenic Copper Accelerator Cavity

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Abstract:

Cryogenic operation of high-gradient radiofrequency (RF) linear accelerators enables higher acceleration gradients and lower breakdown rates, which are critical for compact hadron therapy systems. This study presents a conduction-cooled cryogenic system operating at 77 K for an S-band high-gradient copper traveling-wave accelerator dedicated to ion beam acceleration. The system integrates a custom vacuum cryostat with a Gifford-McMahon cold head providing 250 W cooling capacity at 77 K. To ensure efficient heat transfer, an optimized multi-link design was implemented to minimize thermal contact resistance, supported by detailed thermal analyses. Flexible bellows accommodate cryogenic contraction and cavity alignment, while heaters maintain temperature stability at 77 ± 1 K. Experimental validations, including cooldown processes, steady-state regulation, and high-power RF testing up to 10 MW, confirmed stable 77 K operation with ± 1 K control accuracy under dynamic RF power loading.

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This compact, semi-automated, and industrially scalable platform offers a robust cryogenic engineering solution for the next generation of compact linear particle accelerators based on cryogenic copper cavities.

Keywords: Conduction cooling, Thermal link, RF cavity, Temperature stability, Hadron therapy accelerator.

1. Introduction

Hadron therapy provides precise dose delivery in cancer treatment, but its clinical application is limited by the large size and high cost of conventional accelerator facilities ^[1]. Compact linear accelerators (linacs) are preferred to reduce the facility scale and cost ^[2]. To further shrink the linac size, the development of high-gradient normal-conducting copper accelerating structures is required. Among various frequency bands, S-band technology remains the most mature route to achieve the necessary acceleration gradients for therapeutic linacs.

Most studies on S-band high-gradient copper accelerators focus on room-temperature operation, including geometric design and optimization ^[2], novel fabrication techniques ^[4] and high-power tests ^[5]. Although cryogenic performance of S-band structures has been investigated for electron beams ^[6], cryogenic systems at X- and C-bands have also been developed for compact X-ray free electron lasers ^[7], Colliders ^[8] and RF photoguns ^[10] ; however, these are almost exclusively applied to electron beam. To the best of our knowledge, no cryogenic system operating at 77 K has been engineered for ion beams. Cooling oxygen-free high conductivity (OFHC) copper to cryogenic temperatures decreases surface resistance. This increases the cavity shunt impedance (Z_{TT}) and intrinsic quality factor (Q_0) ^[11], and reduces RF power loss ^[13]. Low temperatures also increase copper hardness, which reduces

surface deformation and raises the RF breakdown threshold to allow higher accelerating gradients^[11]. Motivated by these physical effects, our group previously reported on the design, fabrication, and tuning of a cryogenic high-gradient traveling-wave cavity for a compact carbon-ion therapy accelerator^[15]. However, bridging the gap between a standalone cavity and a fully integrated cryogenic module requires addressing critical engineering challenges, particularly regarding thermal stability under dynamic RF heating and structural integrity during cool down. Based on a comprehensive literature survey, no conduction-cooled cryogenic system operating at 77 K has been reported for an S-band copper traveling-wave accelerator dedicated to ion beam acceleration.

Two methods are typically used to cool high-gradient copper structures: liquid nitrogen (LN) immersion and cryocooler-based conduction cooling. LN immersion^[11] provides good cooling rates and temperature uniformity, but it requires large dewars, transfer lines, phase separators, and complex safety systems, which are not practical for hospitals. Conduction cooling^[16] is simpler and more compact. However, previous conduction cooling designs were mostly built for small test cavities. They cannot handle the dynamic heat load or provide the temperature stability needed for a full-scale S-band traveling-wave accelerator module. In copper cavities, even minor temperature fluctuations cause thermal expansion or contraction, leading to RF resonant frequency detuning and a drop in acceleration performance. To address these challenges, we developed a full-scale conduction-cooled cryogenic system for the accelerator.

In this work, we designed and fabricated an S-band high-gradient traveling-wave accelerator using conduction cooling^[17], and performed its first high-power RF test to verify the cooling scheme. Based on these efforts, this paper presents the development and validation of a cryogenic system, which maintains the accelerator cavity at 77 ± 1 K. Section 2 describes the cryogenic design, including temperature control, sensor placement, and sealing at the cavity-cryostat interfaces. Sections 3 and 4 present the thermal and mechanical analysis. Section 5

details the mechanical structure and assembly. Section 6 covers the cooldown tests, high-power RF tests, and the experimental data analysis.

2. Overview of the Accelerator parameters and Cryogenic system

The S-band high-gradient traveling-wave accelerator cavity ^[15] described in this paper is a negative-harmonic structure operating at 2856 MHz, designed to achieve 60 MV/m at 77 K with 8.2 MW RF input power. Under an RF input power of 10 MW (2 μ S pulse width, 5 Hz repetition rate; hereafter 2 μ S, 5 Hz), the cavity dissipates 58.6W average heat load. Following fabrication and tuning to meet the target electric field distribution, a dedicated conduction-cooling cryogenic system was developed to maintain the cavity at 77 ± 1 K under both no-load and high-power condition.

Although the conduction cooling layout draws on established practices from superconducting RF cavities systems ^[18], the present system operates at significantly higher temperatures (77 K vs. 4.2 K) and must accommodate distinct dynamic heat loads characteristic of normal-conducting copper structures. Accordingly, all mechanical components were redesigned for the specific thermal and operational requirements of this S-band accelerator. Fig. 1 shows a three-quarter cross-sectional view of cryogenic system, which was comprised of five functional subsystems: accelerator cavity, conducting cooling modules, heating modules, two independent vacuum systems, and a temperature regulation system. The primary design objective was to maintain stable 77 ± 1 K operating via cryocooler-based conduction cooling. Flexible thermal links played a critical role in the heat transfer process; their design was required to achieve high thermal conductance, convenient assembly and effective vibration isolation.

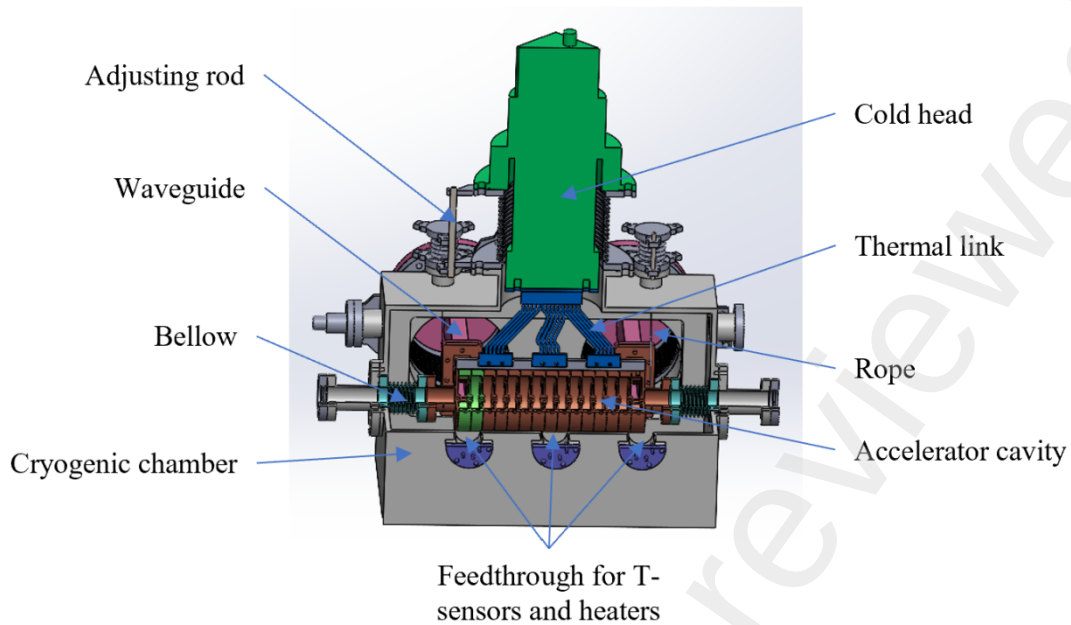


FIG. 1. A three-quarter cross-sectional view of the designed cryogenic system

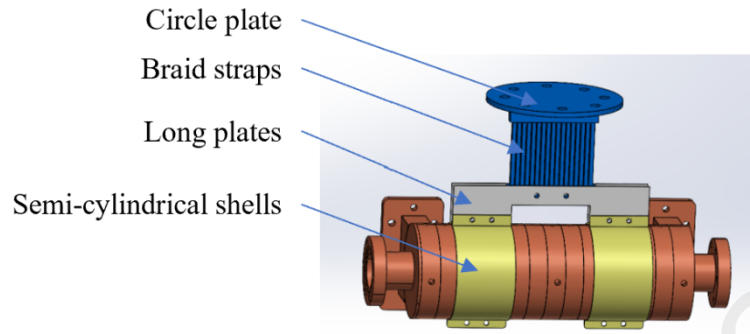
The cavity and cryogenic chamber were equipped with independent vacuum systems to satisfy their respective vacuum requirements. The cavity featured two dedicated RF ports (input and output waveguides) and two beam ports (inlet and outlet). To ensure system reliability, these ports extended externally from the cryogenic chamber to the outer wave guide system and Faraday cup. Maintaining the cavity integrity within a vacuum design target of $< 9 \times 10^{-7}$ Pa, rendered the sealing design of the chamber-accelerator joints a significant technical challenge. To address this, metal-sealed CF flanges were adopted for static interfaces, while edge-welding bellows were integrated at dynamic joints to compensate for differential thermal contraction during cooldown.

Cavity temperature regulation was achieved through a dynamic balance between active cooling (provided by the cryocooler cold head) and active heating (via cartridge heaters). Temperature sensors were strategically deployed to monitor spatial temperature distributions at critical locations during cooldown and steady-state operation. In addition to the aforementioned

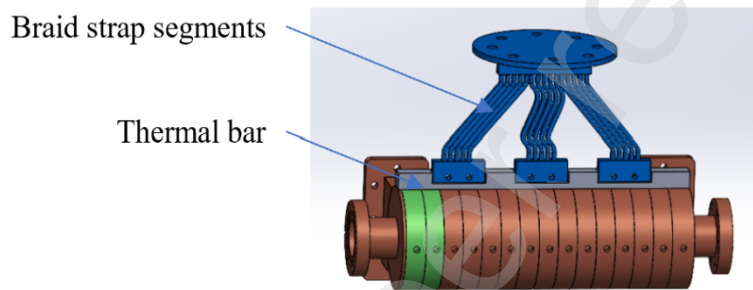
edge-welded bellows for thermal compensation, two suspension ropes were utilized for precise and stable fixation of the cavity within the cryogenic chamber.

3. Thermal analysis

To verify thermal performance, two distinct thermal link design schemes were proposed and comparatively analyzed (Fig. 2). The cavity was precision-machined from oxygen-free high-conductivity copper (OFHC Cu; Residual Resistivity Ratio ≥ 300). Configuration (a) adopted a single-link (OFHC Cu) scheme with indirect mechanical coupling : four 1 mm-thick semi-cylindrical shells were tightly wrapped around the cavity outer surface, and two 1-mm-thick long plates were bolted to connect the shells to the end bar of the thermal link. This link consisted of a 5-mm-thick circle plate, and multiple braid straps with the end bar. The circle plate was rigidly bolted to the cold head baseplate, while the straps end bar was fastened to the long plates via bolted joints. In contrast, configuration (b) employed a distributed multi-link scheme to evaluate the effect of path multiplicity on heat transfer efficiency. A thermal bar (16 mm $\times$ 15 mm cross-section) was integrally brazed to the top of the cavity. This configuration was derived by splitting the overall conductive path of configuration (a) into multiple parallel segments for comparison. While the simulation model of configuration (b) assumed three idealized braid straps per segment, the as-built assembly was upgraded to four segments, each incorporating a double-layer fine copper wire braid architecture. This modification was implemented to provide additional margin against manufacturing tolerances and to achieve higher effective conductance within the same envelope. Consequently, the effective conduction path length and contact resistance per segment in the final application differs from those in the simulation model; specific correlation details are provided in the as-built configuration section (Fig. 8). Each segment was individually bolted to discrete positions on the thermal bar.



(a) Single-link configuration



(b) Multi-link configuration

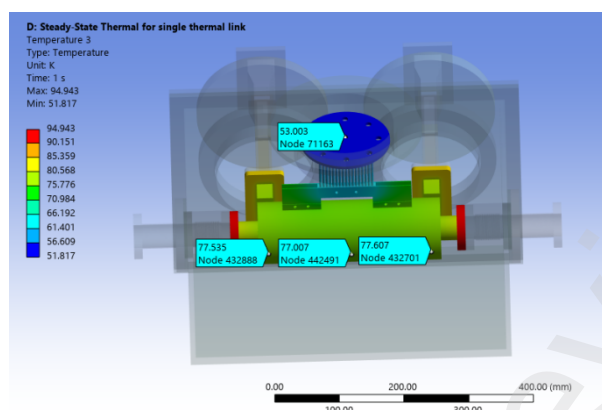
FIG. 2. Heat transfer configurations

Based on these idealized models, thermal performance was evaluated for both configurations using the design models described above, accounting for natural convection on the ambient-exposed outer wall and radiative heat load on the vacuum-facing inner wall of the chamber. Material properties for OFHC Cu and 304 stainless steel were obtained from NIST open data [20]. In the single-link design (Figs. 2(a) and 3(a)), maintaining the cavity at 77 K required a net cooling capacity of 36.0 W at the cold head. Under this load, the copper plate top temperature stabilized at 53 K, resulting in a 24 K temperature drop across the link. This large gradient arose from the high cumulative thermal resistance of the concentrated heat transfer path, which necessitates a substantial driving temperature difference to transport the full cavity heat load to the cold head.

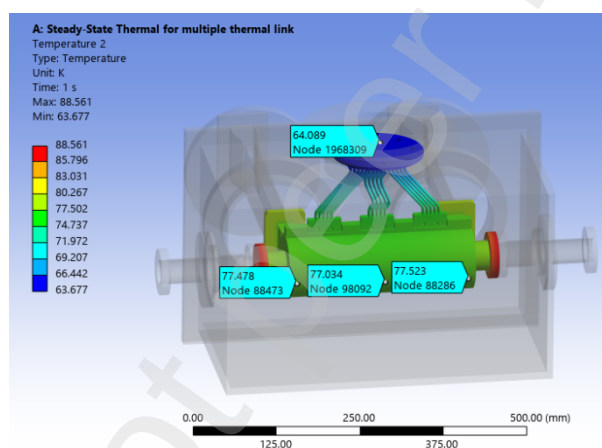
In contrast, the proposed multi-link distributed configuration (Figs. 2(b) and 3(b)) required only 23.2 W of net cooling power to maintain the same 77 K cavity temperature. The copper plate top stabilized at 64 K, reducing the link temperature drop to 13 K. By distributing the thermal load across multiple parallel interfaces, this design significantly lowers the effective thermal resistance, thereby eliminating unnecessary overcooling of the cold head while reliably extracting the intrinsic heat leakage from the cavity.

These results confirm that the multi-link arrangement effectively mitigates the thermal gradient between the cold plate (representing the cold head) and the 77 K cavity. The reduction in required cooling power directly improves system thermal efficiency and temperature stability, providing greater operational controllability and establishing a robust thermal foundation for future accelerator performance upgrades. To further evaluate the dynamic robustness of this configuration under realistic operating conditions, Fig. 4(a) illustrates the simulated thermal response as a function of RF input power (2 μ s, 5 Hz). The equivalent average heat loads were derived from the aforementioned cavity simulation benchmark, which reported 12 W per 2500 ns pulse at 60 MV/m. Scaling this reference to the present pulse parameters yields 48 W at 8.2 MW designed power; linear extrapolation to the 10 MW as an upper-bound test condition gives 58.6W. These thermal loads, evaluated at 2MW peak-power increments, were superimposed onto the zero-power net cooling baseline. Under this coupled model, the mid-section cavity temperature increased by only 0.3 K (from 77.0 K at zero power to 77.3 K at full load), demonstrating exceptional thermal stability across the entire power sweep. Crucially, both the cold head temperature (T_{CH}) and net cooling power (P_{NC}) exhibited linear dependencies on the applied RF input power (P_{RF}): $T_{CH} = -2.17 \cdot P_{RF} + 63.56$. Notably, P_{NC} increased at a rate of 5.86 W/MW, directly quantifying the cavity effective thermal load. Fig. 4(b) details the axial temperature distribution under the 10 MW full-load condition. At this point, the cold head should be delivers a total net cooling power of 81.8 W (sum of the 58.6 W RF heat load and

23.2 W baseline cooling), confirming uniform thermal management along the cavity axis. This simulated linear relationship serves as the baseline for experimental validation in Fig. 14.

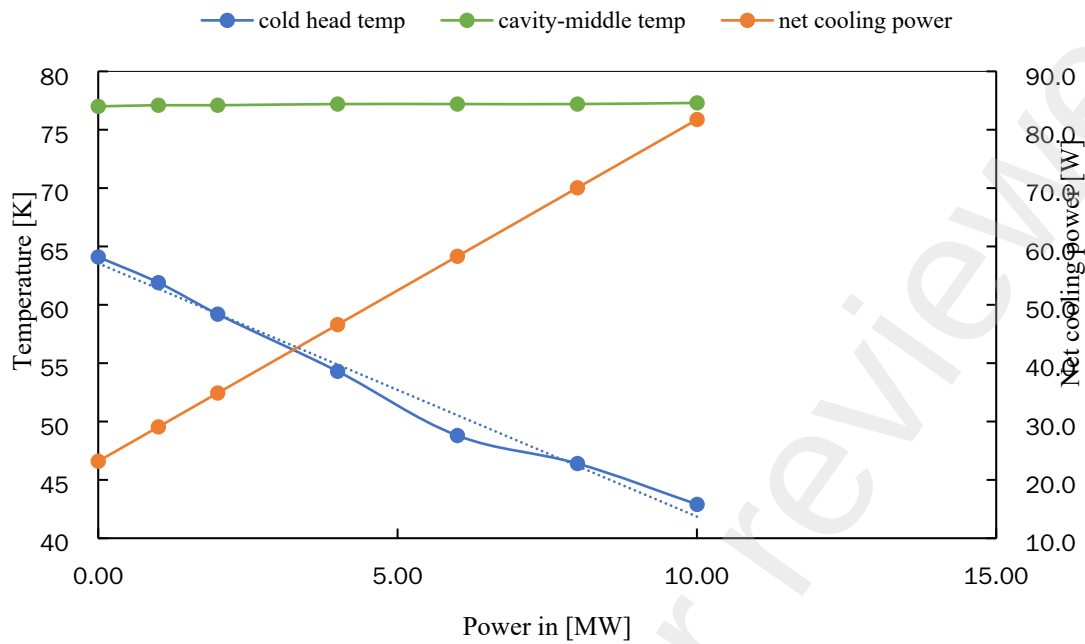


(a) Single-link configuration result

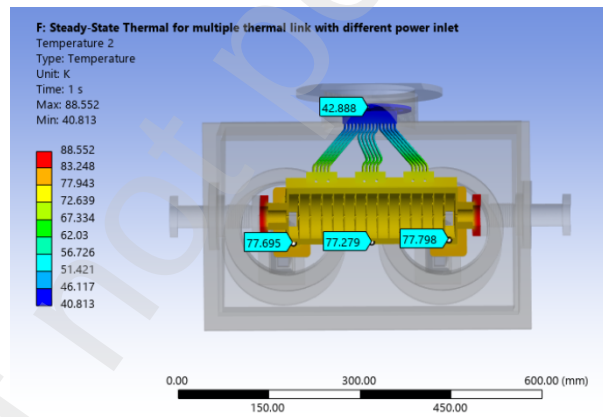


(b) Multi-link configuration result

FIG. 3. Heat transfer results without RF power



(a) Cold head temperature, cavity mid temp and net cooling power as a function of peak RF input power. The cavity-middle temp remains stable at 77.0 K-77.3 K.



(b) Axial temperature distribution along the cavity (inlet, middle, and outlet) under the full 10 MW peak load, corresponding to a net cooling power of 81.8 W

Fig. 4. Simulated thermal response under pulsed RF loading ($2 \mu\text{s}$, 5 Hz).

4. Structure analysis

Following the preliminary design, a structural analysis of the cryogenic chamber was conducted to validate its mechanical integrity under vacuum and gravitational loads (Fig. 5).

The assembly was optimized into a compact configuration with overall dimensions of 600 mm $\times$ 300 mm $\times$ 300 mm. The chamber was fabricated from 316L stainless steel (yield strength: 170 MPa) ^[19] with a 10 mm wall thickness; reinforcing ribs were welded on the front and bottom surfaces to enhance stiffness. The applied load comprised atmospheric pressure (1 atm) and gravitational weight of related components: cold head (18 kg), cavity (13.4 kg), and bellows (7 kg). With a safety factor 1.5, the allowable stress and deformation limits were set at 113 MPa and 1.5 mm, respectively ^[22]. Finite element analysis results indicated that the maximum equivalent stress was 10.4 MPa and the maximum deformation is approximately 0.02 mm, both satisfying the design criteria with adequate margin.

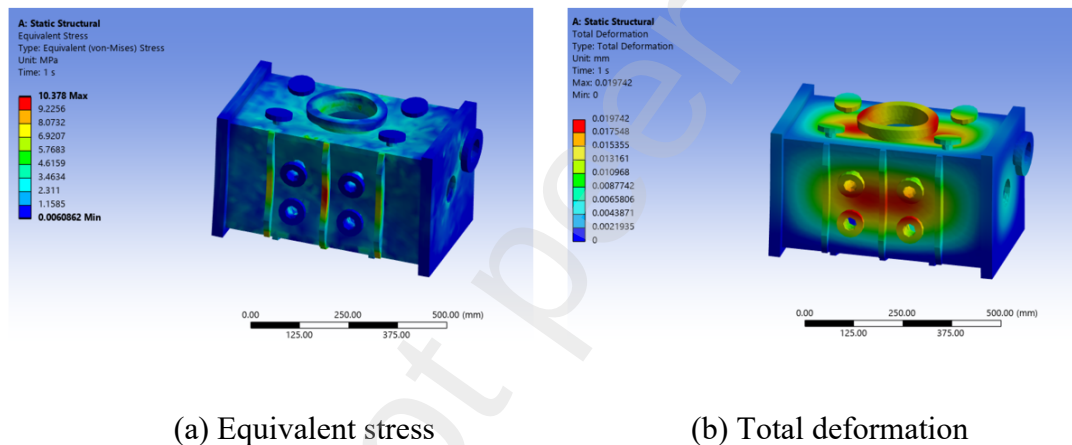


FIG. 5. Structure analysis results of the cryogenic chamber

Upon cooldown from 298 K to 77 K, the accelerator assembly contracts axially. To accommodate this thermal strain without inducing localized stress concentrations, a pair of metallic bellows was integrated as compliant elements, while adjustable support ropes enable fine adjustment of the axial preload. As shown in Fig. 6, the resulting deformation is distributed uniformly along the cavity axis, with a maximum axial contraction of 0.35 mm. Frequency analysis confirmed that the deformed geometry remained within the operational tolerance, with

a frequency shift of only 0.03 MHz with respect to the liquid-nitrogen-immersed reference, well below the prescribed stability limit.

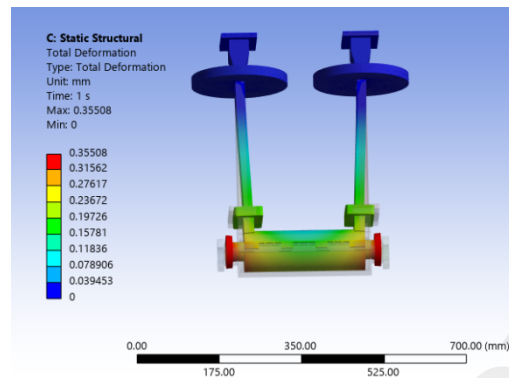


FIG. 6. Total deformation distribution of the cavity upon cooldown from ambient to 77

K

5. System assembly

The assembly sequence of the cryogenic system is illustrated in Fig. 7. First, the cryogenic chamber and its supporting platform were positioned (Fig. 7(a)). The chamber was adjusted along the x, y, and z axes - coarsely via the platform and finely adjusted relative to the waveguide system - to achieve the alignment within 1 mm. Second, the accelerator cavity and custom waveguide were pre-assembled as an integrated subassembly (Fig. 7(b)). Third, the cooling-heating subsystem (CHS) - comprising a cold head (KDE300SA, 250W @ 77 K), four flexible OFHC braid straps (5 mm $\times$ 45mm cross-section area, which is slightly different with former design), three cartridge heaters (HT-100, 100 W each), and bellows assemblies - was independently pre-assembled and functionally validated (Figs. 7(c) and Fig. 8). Subsequently, the CHS was integrated onto the chamber to isolate vibration (Fig. 7(d)) and its flexible straps were connected to the cavity. The cavity was suspended using two loop-slung 304 stainless steel wire ropes ($\varnothing$ 2 mm) configured in loops, while four small bellows above the chamber enabled fine vertical positioning of the cavity (Fig. 7(e)). Additionally, two slightly

larger bellows with custom sealing flanges were connected to both cavity beam flanges to maintain vacuum integrity while accommodating thermal contraction. Eight temperature sensors, electrical feedthroughs and ceramic viewport were installed, and all sensors and heaters were functionally tested to ensure their reliability before the chamber closure. Finally, the chamber was hermetically sealed with two large cover plates via elastomer O-rings (Fig. 6(f)). All sealing interfaces were verified by helium mass spectrometer leak detector. Fig. 9 shows the fully assembled cryogenic system.

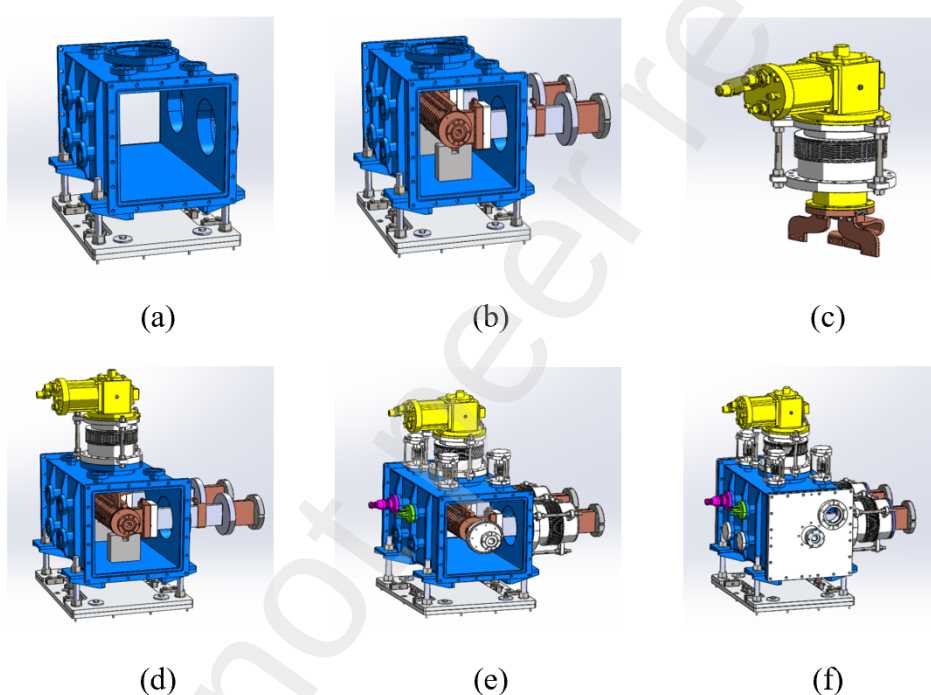


FIG. 7. Cryogenic system assembly sequence. (a) Installation of the chamber on its supporting platform. (b) Pre-assembly of the cavity and the custom waveguide. (c) Pre-assembly and functional validation of the CHS. (d) Integration of the CHS onto the chamber and cavity. (e) Installation of the wire ropes, and bellows for cavity suspension and positioning. (f) Sealing with the cover plates.

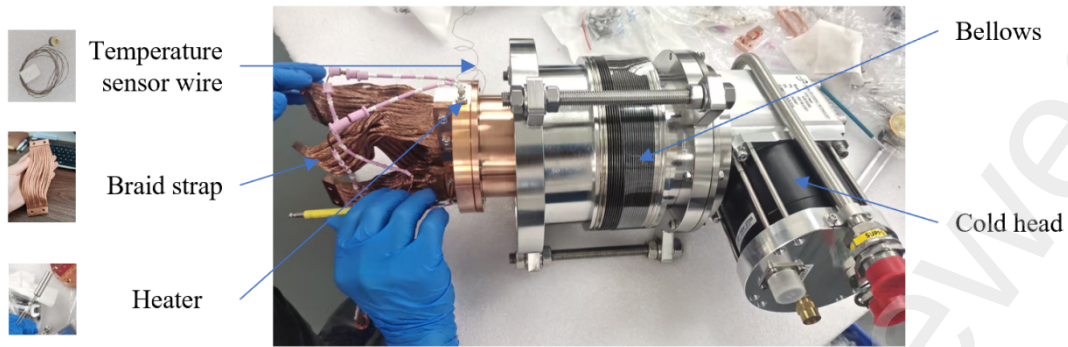


FIG. 8. Photograph of the fully assembled CHS

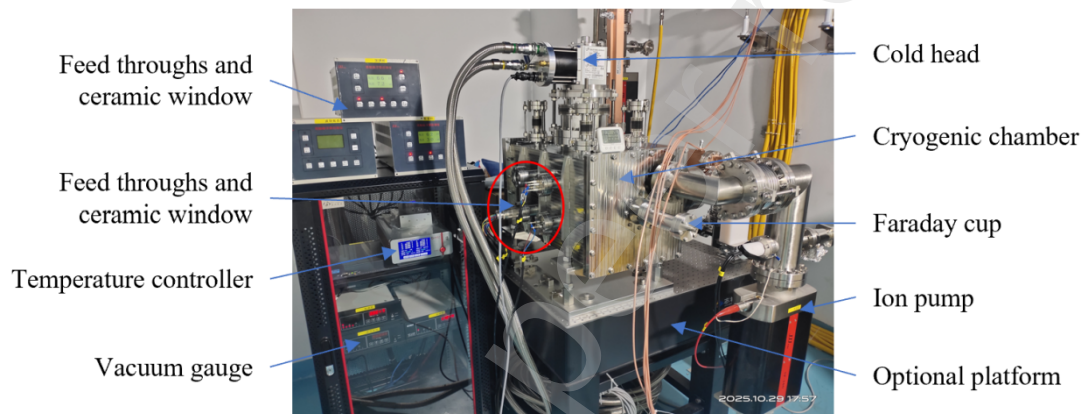


FIG. 9. Photograph of the fully assembled cryogenic system

6. Cooling test

After assembly and leakage testing finished, the integrated system was subsequently evacuated by the pumping system. The cavity sealed with copper gaskets achieved a vacuum level of 2×10^{-7} Pa, whereas the cryogenic chamber sealed with elastomer O-rings and copper gaskets reached 1.5×10^{-5} Pa at room temperature; this discrepancy is attributed to the higher permeation and outgassing rates of the elastomeric seals. Subsequently, cooldown tests were performed. Prior to activating the helium compressor (KDC6000), the water chiller was started to pre-cool the circulating water to 15 °C. The high-pressure helium gas (~ 2.3 MPa) from the

compressor was delivered to the cold head; after heat exchange, the low-pressure helium gas (~ 1.9 MPa) returned to the compressor for re-cooling.

In this test, eight PT100-BO platinum resistance temperature sensors (range: 20-325 K, accuracy: ± 0.1 K) were installed at critical locations (Fig. 10): (1) three sensors (S1-S3) spaced on the cavity bottom surface to assess radial temperature uniformity; (2) one sensor (S4) on the copper plate base to monitor the cold head cooling performance; (3) two sensors (S5-S6) fixed on the cavity flange (adjacent to the waveguide interface) to detect the thermal isolation effect of the stainless steel flange; (4) another two sensors (S7-S8) installed on the waveguide sealing plate base to verify the internal chamber temperature.

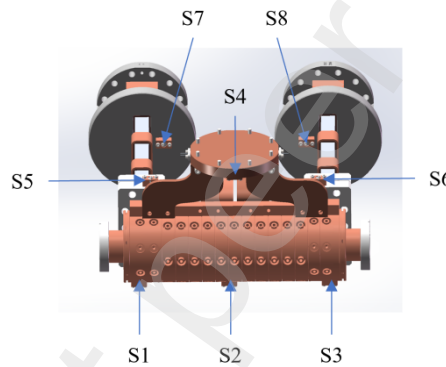


FIG. 10. Distribution of the eight temperature sensors in the cryogenic system

This cooling test reduced the cold head temperature from 298 K to ~ 30 K, and the cavity temperature to ~ 70 K. Throughout this process, the flange temperature was approximately 20 K higher than that of the cavity, which indicated a good thermal isolation. The chamber inner wall temperature decreased by only ~ 3 K due to natural convection. To stabilize the cavity at 77 K, three cartridge heaters were deployed: two connected in series (providing up to 200 W total with adjustable current), while the third was regulated via the supplier's default PID Settings. Through manual iterative adjustments based on temperature sensor feedback, an optimal control strategy minimizing cooldown time was established:

Step 1: Within 2.5 h, the cavity was cooled from 298 K to 77 K while the cold head temperature reached 42 K. During this phase, a 100 L/s ion pump significantly improved the cryogenic chamber vacuum: the cavity pressure stabilized at $\sim 1.8 \times 10^{-7}$ Pa, and chamber pressure decreased from 2.5×10^{-5} Pa to 5.4×10^{-6} Pa.

Step 2: Over the subsequent 0.5 h, the cold head temperature was raised to 68.8-69 K (for seasonal heat leakage variations), while maintaining the cavity at 77 K (as detailed in Figs. 11 and 12).

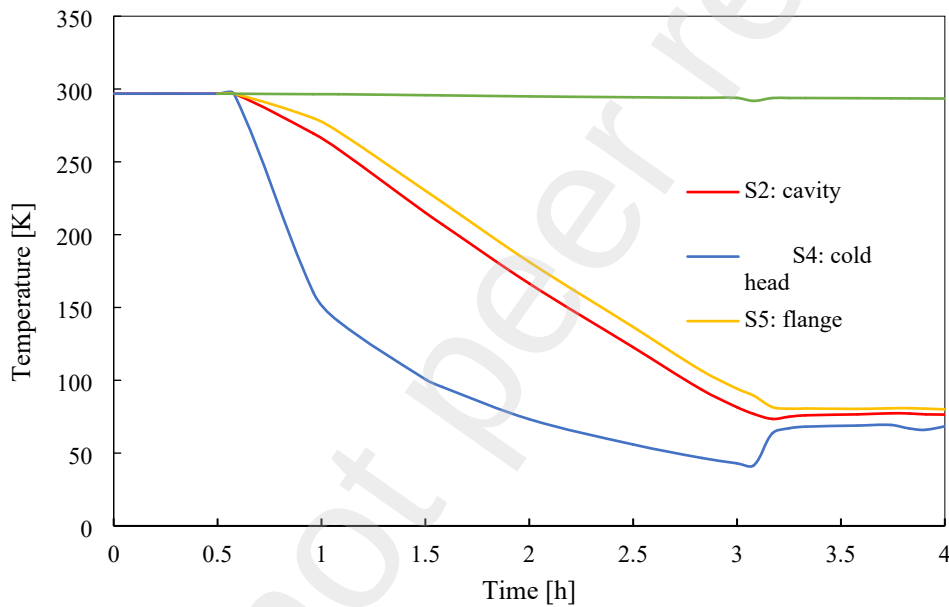


FIG. 11. Cavity cooldown from ambient to 77 K. Temperature evolution of the cavity, cold head, flange, and chamber during cooldown. Thermal equilibrium is reached at ~ 77 K after ~ 3 h.

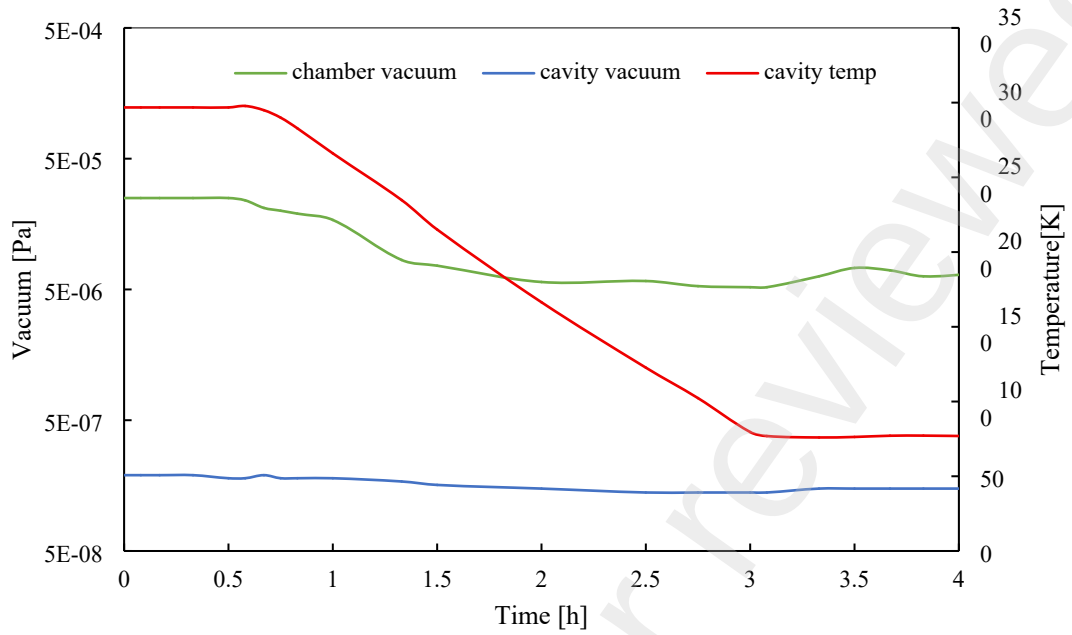


FIG. 12. Cryogenic chamber and cavity vacuum change with cavity temp decrease

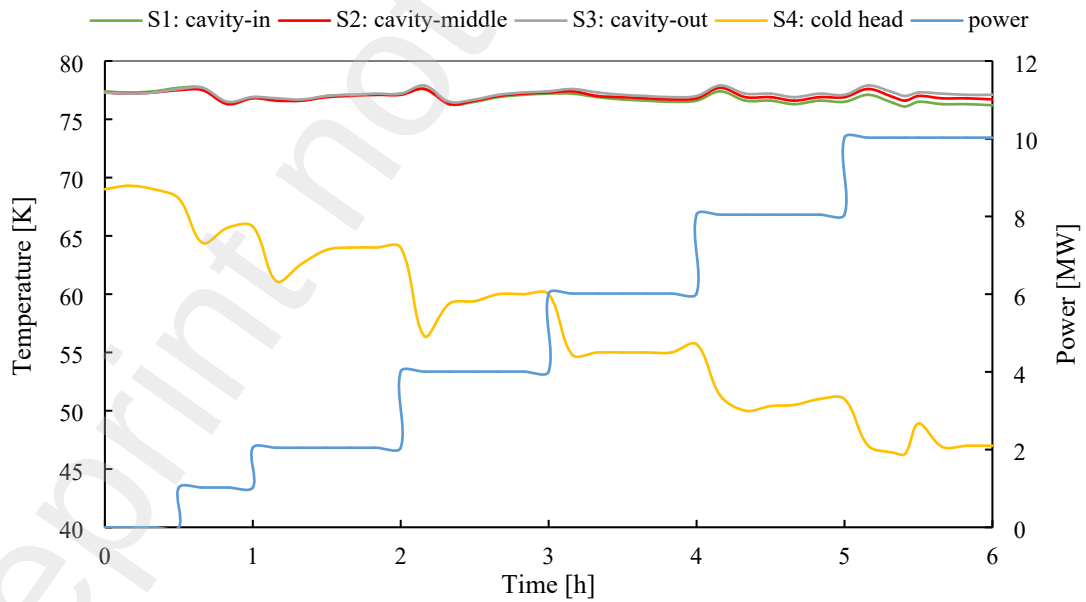


FIG. 13. Time-dependent cold head temperature (T_{CH}) and cavity temperature (T_{CAV}) during RF power step tests. The cavity temperature is actively stabilized at 77 ± 1 K throughout the measurement.

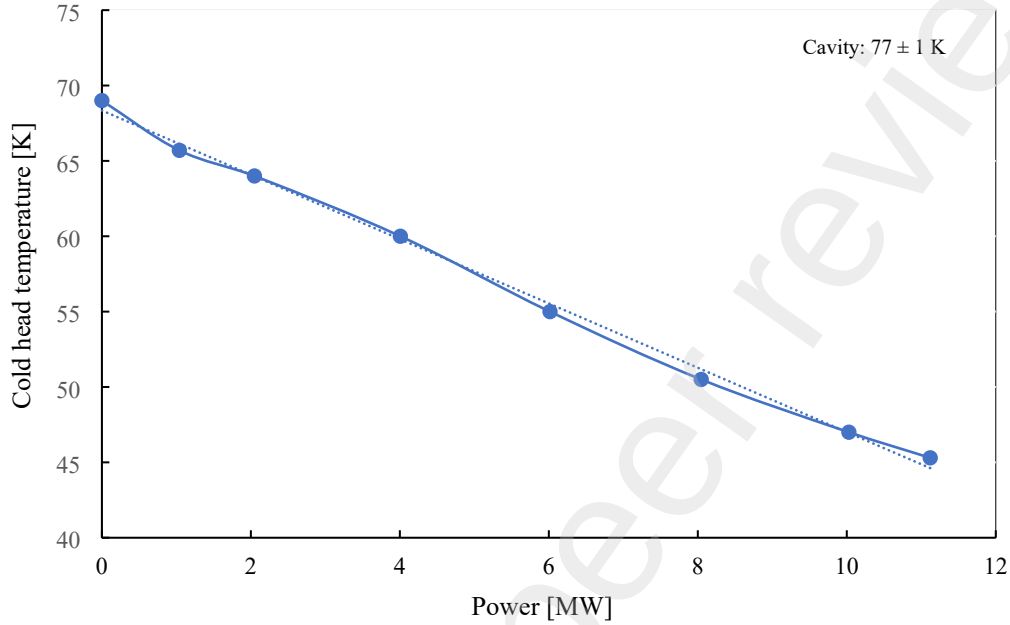


FIG. 14. Steady-state cold head temperature (T_{CH}) versus RF input power (P_{RF}) during steady-state operation with the cavity maintained at 77 ± 1 K. Data points represent 10-min averaged values extracted from Fig. 13. The dashed line represents an empirical linear fit ($T_{CH} = -2.13 \cdot P_{RF} + 68.35$, $R^2 = 0.997$) intended solely as a practical calibration reference.

Stable cavity temperature is a prerequisite for high-power load testing. During the initial two-week conditioning phase (0-11.2 MW, 1-3 μs pulse width, 5 Hz repetition rate), intermittent flashovers in the cavity and waveguide system caused transient vacuum degradation and temperature excursions, both of which subsequently self-recovered. After approximately one week of continuous operation at the peak power of 11.2 MW (2 μs , 5 Hz),

flashover frequency diminished to an acceptable level, enabling the formal load test described in Fig.13.

Under this steady-state test protocol, RF power was stepped from 0 MW to 10 MW (2 μ s, 5 Hz) with dwell times of 30 min at 0 MW and 1 MW, followed by 1 hour at each of 2, 4, 6, 8, and 10 MW. During the zero-RF-power baseline period, the measured temperature drop across the as-built four-segment thermal link was 8 K, confirming that the fabricated assembly provides sufficient intrinsic thermal conductance to meet the design requirement for stable 77 K operation under the actual experimental conditions. As RF power increased, the cold head temperature decreased from 69 K to 47 K, while the cavity temperature remained stable at 77 ± 1 K throughout the entire power sweep. At the maximum power level of 10 MW, the accelerating gradient reached 66 MV/m. These results demonstrate effective cavity temperature regulation under dynamic RF loading. Furthermore, Fig. 14 shows a clear linear trend between RF input power and cold head temperature under stable cavity operation (77 ± 1) K. The empirical linear relationship ($T_{CH} = -2.13 \cdot P_{RF} + 68.35$) provides a practical calibration tool: operators can anticipate the cold head temperature shift based on RF power settings, enabling proactive thermal management without requiring knowledge of absolute heat loads. Compared with the simulation baseline established in Fig. 4, the experimental data exhibits a highly consistent linear slope, confirming that the thermal load dynamics were accurately predicted. Notably, the measured cold head temperatures are consistently higher across the entire power range, showing a parallel upward shift of approximately 4.79 K. This offset is attributed to the geometric and contact resistance differences between the actual strap used in the assembly and the idealized strap modeled in the simulation, which slightly increased the overall thermal resistance.

7. Summary and Outlook

This study established a validated engineering reference for a 77 K conduction-cooled cryogenic system designed for an S-band high-gradient copper accelerator cavity, specially tailored for compact ion accelerators. While prior works have explored cryogenic cavity structures in isolation, the present contribution lies in the integrated system-level validation of a cold head, a multi-link configuration, heaters, bellows and ropes. Thermal simulations evaluated heat transfer efficiency during cooldown from ambient to 77 K under no-load conditions for two candidate configurations, based on which the preferred design was selected for subsequent RF-powered thermal analysis. Parallel structural analysis informed design refinements to mitigate thermally induced deformation during this cooling phase. Guided by these design and simulation results, precision assembly procedures were implemented to minimize interfacial thermal resistance and ensure mechanical alignment. Subsequent cooling tests validated the simulated cooldown performance and demonstrated stable cavity temperature regulation at 77 ± 1 K under RF loads up to 10 MW (66 MV/m). Collectively, this integrated workflow establishes a verified engineering basis for conduction-cooled high-gradient copper accelerator cavity, enabling industrial-scale development. Future efforts will focus on: (1) assessing long-term reliability under sustained high-power cycling; and (2) refining autonomous temperature regulation strategies for unattended operation. These developments aim to bridge the gap between laboratory validation and clinical translation.

CRedit authorship contribution statement

Min Yang: Writing - review & editing, writing - original draft, validation, investigation, analysis, data curation. **Quantang Zhao:** Writing - review, supervision, investigation, conceptualization, funding acquisition. **Peng Yu:** Validation, investigation. **Ruikang Ma :** Validation, and Remote-control program interface installation. **Chihao Ni:** Validation.

Xiaoxiao Yuan: Validation. **Jiao Xu:** Validation. **Hongwei Zhao:** Funding acquisition, conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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